

MUD LOGGING GEOLOGICAL EVALUATION & WELL CONTROL

SYNTHETIC DEMONSTRATION DOCUMENT - NOT REAL OPERATIONAL DATA (SIH PROTOTYPE)

WELL IDENTIFIER: NWIS-W003
FORMATION: Demo-Tipam Sandstone
EVENT TYPE: KICK

OPERATIONAL DATE: 2024-05-18
MEASURED DEPTH: 2240.0 m
SEVERITY LEVEL: HIGH

LITHOLOGICAL EVALUATION

Drilling 12-1/4 inch hole section in Demo-Tipam Sandstone (1620m - 2480m).
Clean, porous, sub-angular to sub-rounded quartzose sandstone. Visible porosity 22-26%.
At 2240m, sudden drilling break observed (ROP jumped from 9 m/hr to 26 m/hr).

WELL CONTROL INCIDENT

Flow check performed: positive flow detected with mud pumps off. Pit gain of 18 bbls recorded.
Encountered abnormal high-pressure gas pocket charging permeable Tipam sandstone lenses.
Total gas on mud log detector peaked at 48% with significant methane (C1) and ethane (C2) readings.

WELL KILL MITIGATION

1. Shut in well on annular preventer; recorded initial pressures: SIDPP = 340 psi, SICP = 460 psi.
2. Formulated kill mud calculation: raised required mud weight from 1.18 SG to 1.30 SG.
3. Executed Wait-and-Weight kill method; circulated gas kick bubble out safely via choke manifold.
4. Verified zero trapped pressure; resumed controlled drilling operations.